

Zehao Dong

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Profile

I am a PhD candidate in Physics at Tsinghua University, focusing on strongly correlated quantum materials and advanced electron microscopy. I develop GPU-accelerated multislice electron ptychography (MEP) to visualize atomic-scale defects and dopants in three dimensions.

Research Interests

- High-temperature superconductivity in cuprates and nickelates;
- Scanning transmission electron microscopy at atomic resolution
- Multislice electron ptychography; GPU-accelerated computational imaging and phase retrieval

Education

Tsinghua University, PhD in Physics 2022 – present

- Research: Electron ptychography (MEP), 4D-STEM, strongly correlated materials (nickelates/cuprates)
- Advisors: Prof. Yayu Wang; Prof. Zhen Chen

Peking University, BSc in Physics 2018 – 2022

- GPA: 3.84/4.00
- Advisor: Prof. Yu Ye

Selected Publications

- **Interstitial oxygen order and its competition with superconductivity in $\text{La}_2\text{PrNi}_2\text{O}_{7+\delta}$** 2025
Zehao Dong[†], Gang Wang[†], Ningning Wang[†], Wen-Han Dong[†], Lin Gu, Yong Xu, Jinguang Cheng*, Zhen Chen* & Yayu Wang*
 Nature Materials, 24, 1927–1934 (2025)
- **Antiferromagnetic quantum anomalous Hall effect under spin flips and flops** 2025
 Zichen Lian[†], Yongchao Wang[†], Yongqian Wang, Wen-Han Dong, Yang Feng, *Zehao Dong*, Mangyuan Ma, Shuai Yang, Liangcai Xu, Yaoxin Li, Bohan Fu, Yuetan Li, Wanjun Jiang, Yong Xu, Chang Liu*, Jinsong Zhang* & Yayu Wang*
 Nature, 641, 70-75 (2025)
- **Sub-nanometer depth resolution and single dopant visualization achieved by tilt-coupled multislice electron ptychography** 2025
Zehao Dong, Yang Zhang, Chun-Chien Chiu, Sicheng Lu, Jianbing Zhang, Yu-Chen Liu, Suyu Liu, Jan-Chi Yang, Pu Yu, Yayu Wang & Zhen Chen*
 Nature Communications, 16, 1219 (2025)
- **Visualization of oxygen vacancies and self-doped ligand holes in $\text{La}_3\text{Ni}_2\text{O}_{7-\delta}$** 2024
Zehao Dong[†], Mengwu Huo[†], Jie Li[†], Jingyuan Li, Pengcheng Li, Hualei Sun, Lin Gu, Yi Lu*, Meng Wang*, Yayu Wang* & Zhen Chen*
 Nature, 630, 847-852 (2024)
- **The emergence of global phase coherence from local pairing in underdoped cuprates** 2023
 Shusen Ye, Changwei Zou, Hongtao Yan, Yu Ji, Miao Xu, *Zehao Dong*, Yiwen Chen, Xingjiang Zhou & Yayu Wang*
 Nature Physics, 19, 1301-1307 (2023)

(For a complete and up-to-date list, please see my Google Scholar)

Research Experience

PhD Research, Tsinghua University, Department of Physics 2022 – present

- Developed GPU-accelerated multislice electron ptychography (MEP) pipelines.
- Introduced tilt-coupled MEP for sub-nanometer depth resolution and single-dopant 3D visualization.
- Applied 4D-STEM/MEP to nickelate superconductors to quantify oxygen vacancies, ligand holes, and interstitial oxygen ordering.
- Tools: MATLAB, Python; NVIDIA A100/RTX 4090 clusters; STEM and EELS.

Undergraduate Research, Peking University, School of Physics 2019 – 2022

- Fabricated ALD-grown AlO_x planar tunnel junctions for vdW superconductors.
- Low- T spectroscopy and angular magnetotransport.
- STM/tunneling spectroscopy on cuprates and conventional superconductors.

Fundings & Awards

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|---|-------------|
| • NSFC's Young Scientists Fund for Graduate Students (Grant No. 124B2068) | 2025 – 2026 |
| • ICAM-China Emerging Scientist Award | 2025 |
| • Best Poster Award, International Conference on the Fundamentals of Quantum Matter | 2025 |
| • National Scholarship for Graduate Students, Ministry of Education, China | 2024 & 2025 |
| • Best Oral Presentation Award, 20th International Microscopy Congress | 2023 |
| • Gold Medalist, 49th International Physics Olympiad (IPhO) | 2018 |

Conference Presentations

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| • International Conference on the Fundamentals of Quantum Matter, Beijing, China | 2025 (Poster) |
| <i>Visualizing oxygen defects that govern superconductivity in bilayer nickelates via electron ptychography</i> | |
| • CPS Fall Meeting, Harbin, China | 2025 (Poster) |
| <i>Interstitial oxygen order and its competition with superconductivity in $\text{La}_2\text{PrNi}_2\text{O}_{7+\delta}$</i> | |
| • CPS Fall Meeting, Haikou, China | 2024 (Oral) |
| <i>Visualization of oxygen vacancies and self-doped ligand holes in $\text{La}_3\text{Ni}_2\text{O}_{7-\delta}$</i> | |
| • Advanced Transmission Electron Microscopy Conference, Hong Kong, China | 2024 (Poster) |
| <i>Visualization of oxygen vacancies and self-doped ligand holes in $\text{La}_3\text{Ni}_2\text{O}_{7-\delta}$</i> | |
| • 20th International Microscopy Congress, Busan, South Korea | 2023 (Oral) |
| <i>Improving depth resolution using tilt-series coupled multislice electron ptychography</i> | |

Technical Skills

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- **Experimental:** STEM/4D-STEM, electron ptychography, EELS;
 - **Programming:** MATLAB, Python; GPU computing
 - **Languages:** Chinese (Native), English (Fluent)